

SIMULATION OF WATER PERMEATION IN A CNT MEMBRANE

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The objective is to perform a nonequilibrium MD simulation of water permeation across a CNT membrane under applied pressure, starting from a given initial configuration. This is related (but not equal) to the official NAMD tutorial¹ “*Simulation of water permeation through Nanotubes*”.

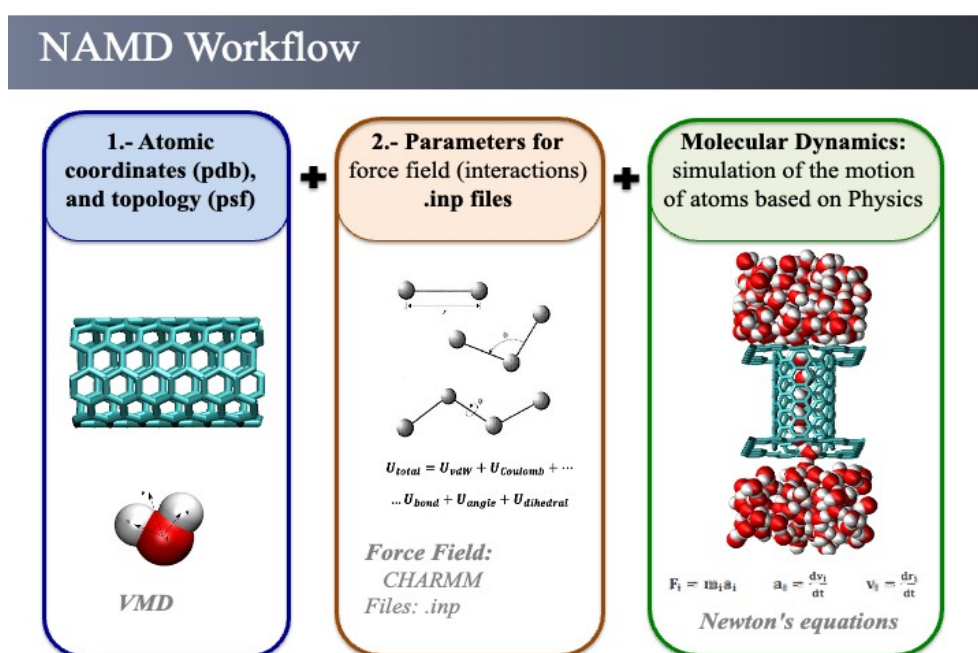
Materials:

- System data for the exercise (Exercise folder) and other NAMD examples of simulations :

https://github.com/jfaraudo/Examples_NAMD

- Github with files to perform and analyse NAMD simulations:

https://github.com/jfaraudo/Running_NAMD



In order to perform a simulation with NAMD we will need the coordinate (pdb), structure (psf) and Force Field (inp) files containing *all the information of the model to be simulated*:

- **A single coordinate file** in pdb format containing initial atomic coordinates for the system
- **A single structure file** in psf format containing: type, mass and charge for each individual atom and a list of all covalent bonds of the system (including bonds, angles and dihedrals)
- **Force Field parameter files** (par_XXX.inp or par_XXX.prm files) in CHARMM or AMBER formats including all parameters of the force field such as force constants of bonds, angles and dihedrals, equilibrium values of bond distances, angles and dihedrals and Lennard-Jones parameters for all atom types. Force field files for NAMD in CHARMM format are available here: https://github.com/jfaraudo/Running_NAMD

¹<http://www.ks.uiuc.edu/Training/Tutorials/namd-index.html>

1. Preparation:

1.1 Create a new folder (a suitable name could be “simulation/input”). Download the pdb and psf system files from the “Exercise” folder in the Examples_NAMD GitHub repository.

1.2 Visualize the system with VMD. Be sure that you understand the contents. Check that the dimensions of the system correspond to those described at the GitHub readme file.

1.3 Open the pdb file with a text editor. You will see that there is a column (the so-called “beta” field) that is always zero for water and 1 for carbons. Why? (hint: interactive simulation session of water in a nanotube). In VMD the value of “beta” can be easily visualized by going to Graphics – Representations (Graphical Representations window) and for the selection you have (for example, the entire system) select “beta” for the colouring method.

1.4 Download the Force field files (inp) for NAMD in CHARMM format needed for this system:

https://github.com/jfaraudo/Running_NAMD

Be sure to have in your input folder parameter files for the CNTs and for the TIP3P model of water.

3. Energy minimization

In general, coordinates obtained from experimental data or containing solvent added by automated procedures are not suitable for direct MD simulation (they may contain atomic positions affected by errors and/or unphysical configurations). Before running the actual MD simulation of the protein, we need to correct them. The way to do this is by **energy minimization or geometry optimization**. We have two options for this:

- a) Run an interactive NAMD simulation (IMD) as in the previous sessions (protein and nanotube)
- b) Submit the calculation in background from the terminal (as we did previously with SIESTA).

Here we will do option (b) which is the usual one with calculations involving a large number of atoms. To this end, we will need a *NAMD configuration file including all details of the calculation* (this is typical for all simulation programs). The NAMD configuration file can be created either using VMD (Extensions – Simulation – NAMD Graphical Interface) or modifying a previous file available from a tutorial or example. Example configuration files can be obtained at:

https://github.com/jfaraudo/Running_NAMD

3.1 Download the example Energy minimization configuration file in a new folder (a suitable name could be “simulation/minimize”). All files from that calculation will be stored here.

3.2 Open the file with a text editor and **modify it to adapt to your simulation**. In particular always:

- check the names of input files and output files.
- introduce a size of the *simulation box* (or simulation cell or supercell), compatible with your system. In our case, it must correspond to the size of the membrane and water box.
- Check that the presence of fixed (not moving) atoms will be considered.

3.3 In a terminal execute the energy minimization in the background, using all your available processors. Check first the number of processors of your machine (in linux: *lscpu*). Send the namd3 calculation. For example for a 12 processor machine:

```
nohup namd3 +p12 minimization.namd > minimization.log &
```

3.4 Follow the evolution of the calculation from the terminal with the *top* command. Also you can use other terminal commands such as “tail” or “grep” to check the changes in the log file.

Now we will analyse the results and verify whether the energy minimisation was correctly done

3.5 Once the simulation has finished, look at the evolution of the system. Open the coordinate file (dcd file) with VMD and look at the changes. Verify whether the size of the simulation box was correct or not (use “Periodic” in the “Graphical Representations” option).

Also verify the changes in the Potential Energy, which can be extracted and plotted from the log file using the VMD option Extensions – Analysis - NAMD Plot.

4. Run a system equilibration at desired T,p conditions

Once we have an optimized (energy minimized) structure, we can perform the actual MD simulation. Typically we perform first a short run to **equilibrate** the system at the desired conditions (with a thermostat to equilibrate temperature, for example) and from the results we then perform a subsequent **production run**. To perform the simulation we will follow a protocol equivalent to the one used for the energy minimization.

4.1 Download a system equilibration namd configuration file from the same GitHub repository as before. Here it must be a file for NpT conditions **with the pressure applied only in the z direction** without modifying the system volume in the x,y directions (why???).

4.2 Put this file in a new folder (a suitable name could be “simulation/equilibration”). All files from that calculation will be stored here. *Check and modify the file*, in particular:

- check the names of input files and output files.
- check the *time step* value for the MD simulation
- check the *total simulation* time (1 ns should be enough)
- check the *Thermodynamic conditions* (temperature and pressure in this case).
- check the *frequency* of the output data
- Check that the presence of fixed (not moving) atoms will be considered.

4.2 Execute the calculation in background and follow the evolution of the calculations as before.

Now we will analyse the results and verify whether the system is at thermodynamic equilibrium

4.3 As before look at the trajectory (dcd file) with VMD. Do you see any significant change during the simulation? Look also at the orientation and motion of water molecules inside the nanotubes.

4.4 Do you observe any change in the size of the simulation box? Check the time evolution of the size of the system (.xst file that can be plotted with *gnuplot*). The final dimensions are available in the .xsc file. Why the system size changes over time? Is the system size equilibrated?

4.5 Check the convergence of the Temperature, the Potential Energy (why we are not interested in the total energy?) and the pressure during the simulation. You can use VMD as we did before or you can use a jupyter notebook that extracts the relevant data from the log file (see python folder at the GitHub). How much time is needed to reach the thermodynamic equilibrium state?

Question 1: Analyse the behaviour of water inside the nanotubes during the equilibrium state. In particular use the python scripts to compute: (a) the number of water molecules inside the nanotubes as a function of time, (b) the number of permeations in each direction (from +z to -z and vice versa) and (c) the flow of water molecules. How do you interpret these results?

As we you can see, in this short equilibration simulation we have poor statistics of water crossing/flow across the CNTs. In order to get good statistics we have performed a much longer production run that continues this equilibration simulation, available at the GitHub.

Question 2: Repeat all the previous analysis using the production run.

5. Run a nonequilibrium simulation.

In equilibrium, there is no overall flow of water. In order to have a flow of water, a pressure difference must be applied. This corresponds to a nonequilibrium simulation under external force.

5.1 We start from the equilibrium system at the desired T,p conditions and apply a pressure difference between the two sides of the membrane by applying an external force. At https://github.com/jfaraudo/Running_NAMD you will find a script that allows you to apply external forces, applying a force on the water molecules that are in a given region. Download it to a new folder (simulation/nonequilibrium) and modify the script to perform a short simulation (200 ps could be enough). You can use the force proposed in the example or a different value. **Be sure to keep the CNTs atoms fixed.**

5.2 Execute the calculation in background and follow the evolution of the calculations as before.

Now we will analyse the results and observe the effect of the external force

5.3 Open the trajectory with VMD. Look at the orientation and motion of water molecules inside the nanotubes. Do you observe any difference with the equilibrium case?

5.4 Using the provided python scripts analyse the permeation and flow of water across the CNTs. Which differences do you observe from the previous case?

Create a GitHub **Repository** for your data and save the results obtained in the different steps in different folders.